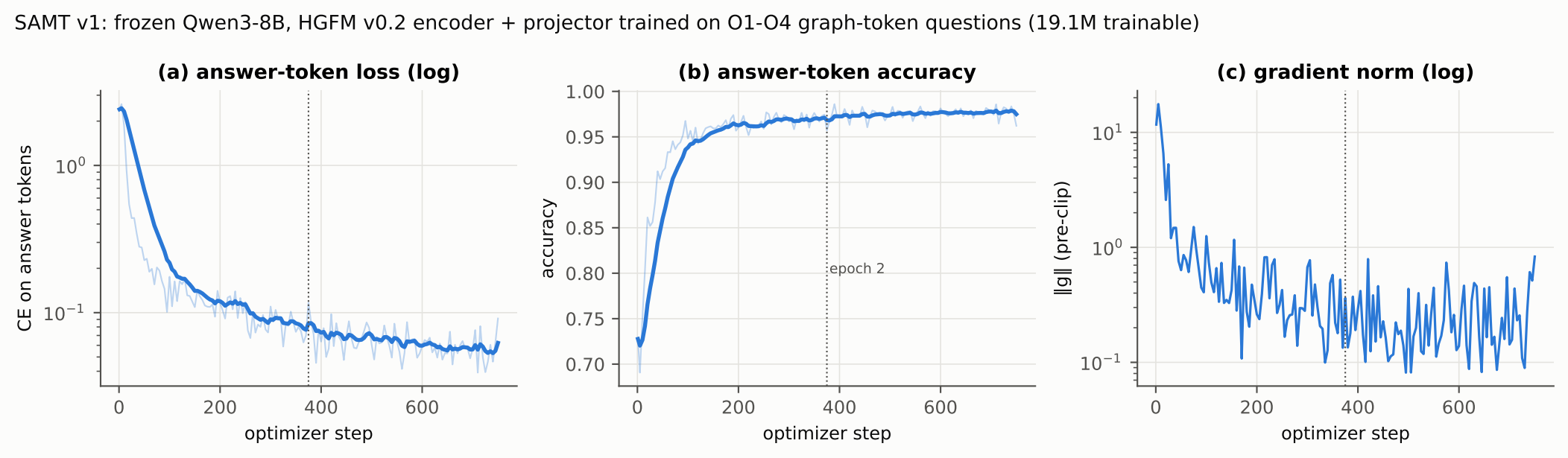
SAMT v1: frozen Qwen3-8B, HGFM v0.2 encoder + projector trained on O1-O4 graph-token questions (19.1M trainable)
(a) answer-token loss (log)
CE on answer tokens
10^{0}
10^{-1}
0
200
400
600
optimizer step
(b) answer-token accuracy
accuracy
1.00
0.95
0.90
0.85
0.80
0.75
0.70
epoch 2
0
200
400
600
optimizer step
(c) gradient norm (log)
‖g‖ (pre-clip)
10^{1}
10^{0}
10^{-1}
0
200
400
600
optimizer step